Ethical Responsibilities of Engineers and Platforms in Prioritizing User Security while Balancing Cost and Service – A Case Study on Telegram

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1. Introduction

1.1 Overview of the Chosen IT Topic

Instant messaging apps are quickly expanding and have changed how we communicate. SMS texting, phone calls, email, are on the decline, and instant messaging apps are on the rise. WhatsApp, Signal and Telegram have quickly become leading communication platforms among billions of users who can share text, images, video, and voice messages. They are also great for group-based and collective and global communities.

Telegram is famous for its speed, ability to host larger groups, and excellent privacy features. Telegram has "Secret Chats" that are end-to-end encryption, vast channels that can accommodate a number of users, and it has a use beyond being just an instant messaging app, being used for activism, political movements, business, and information sharing.

Despite these attributes, Telegram also has significant challenges. One issue is that “Secret Chats” provides end-to-end encryption, while regular chats and groups expect encryption between the devices and the server, which means the company has access to your data in some way. Many users may not understand the difference between Secret Chats and regular chats, and may wrongly assume all of their conversations are completely secure. This scenario is risky, and raises ethical issues of the company's transparency with the users.

This report examines the engineers' and the organizations' ethical responsibilities while communicating on a platform like Telegram, and how the engineers navigate the ethical balance between user privacy, cost, performance, and quality of service. The report also examines if the engineers are fulfilling their obligation to protect users from being hacked.

1.2 Importance of Ethics in IT and Engineering

Information technology transformations occur at a rapid pace, leading to ethical challenges for engineers, designers, and platform operators. For example, in areas like cybersecurity or communications making ethical choices requires balancing technical possibilities with user rights and the needs of the corporate world.[Note 1].

Other than compliance with the law, ethical considerations also encompass protecting the privacy and transparency, as well as avoiding or restricting damage. If these values are abandoned, it can result in serious consequences, that includes data breaches, mass surveillance, and misinformation. Engineers' design decisions have a direct impact on user safety and social trust.

The IEEE and ACM Codes of Ethics are ethical codes that set standards for platforms like Telegram, where user autonomy is prioritized over user welfare due to the combination of technical complexity and mass-scale social interaction.

1.3 Relevance to Society and Technology

Telegram is a very significant part in the digital realm. It has a really diverse user-base. The user-base includes not only activists but also journalists, whistleblowers, and ordinary users in countries where censorship and surveillance are a regular part of life. Telegram is no less than like a lifeline for these people. Because it allows them to speak freely, organize events, share news quickly etc.

At the same time, the openness that makes Telegram helpful can also create problems. Criminal groups, extremists, and people who spread false information often use the platform too. There is another concern, that is, Telegram's rules about encryption and data are not always clear. Lot of users believe their messages are fully secure, but actually they are not. This misunderstanding can put them in real danger.

These issues are actually part of a bigger challenge that all platforms generally face. They have to find a balance between each thing, adding new features, keeping costs low, protecting users, and following government rules. Our report looks at Telegram as an example, that shows why accountability and transparency are important. If platforms like this want to protect the rights, they must act responsibly no matter what and be honest with users.

2. Problem Statement

Telegram faces a very serious challenge: ensuring strong security, maintaining scalable services, and meeting regulatory expectations while managing costs and usability.

The core dilemmas include:

A. <u>Privacy vs. Performance</u>

Telegram only gives full end-to-end encryption in "Secret Chats." Regular chats and groups use client–server encryption instead. This makes the app faster and able to handle very large groups, but it also means the data can be more exposed. Many users do not realize this difference, which creates risks.

B. **<u>Security vs. Cost</u>**

Adding stronger encryption everywhere and moderating harmful content needs a lot of money and technical resources. Telegram has focused more on growth than on protections of user rights. This shows user safety is being traded off for efficiency and lower costs.

C. **<u>Transparency vs. User Awareness</u>**

Telegram shares only limited and highly technical details about its encryption and policies. People cannot easily understand them. So, users think they are safe. It weakens their ability to give proper consent.

D. **<u>Free Speech vs. Harm Prevention</u>**

Telegram's policy of very light moderation allows people to communicate freely. This can be positive in countries with censorship. On the other side, it is really easy for hate speech misinformation to spread. It creates real ethical problems.

These interconnected dilemmas demonstrate how engineering choices shape user trust, safety, and societal impact. Addressing them requires robust ethical frameworks, clear communication, and responsible governance to ensure innovation doesn't compromise user protection.

3. Objectives

This study examines the engineering and governance aspects of the Telegram app in four ways:

Explore the moral aspects associated with security, privacy or business expenditures.

- Observe the design choices that Telegram has made to ensure user privacy, performance quality, and affordability.
- Options include user-friendly end-to-end encryption, client–server data management capabilities, and large group collaboration.
- Explain the ethical obligations of engineers and any gaps in their work performance.

Examine user privacy, security, and trust.

- Take into account realistic scenarios, such as the 2019-2020 season.
- Misinformation, harassment, or data exposure in Hong Kong to link the design of platforms and policies with tangible social impacts.

Propose Ethical Recommendations

The report suggests giving clear advice for engineers, platform owners, and policymakers. These ideas should follow the rules from IEEE and ACM codes of ethics. Besides this, they should use basic ethical ideas like deontology too. The main goal is to guide them to make safe and fair choices when they're building such platforms.

Provide Feedback to Alternative Media Outlets

Share valuable lessons to help other sites and platforms tackle comparable ethical issues, promoting better technology development, better governance, and better digital rights.

4. Literature Review

A foundation for examining the ethical position of Telegram is research across cybersecurity, engineering ethics, and platform governance. Key insights include:

Encryption and User Perceptions:

The mismatch between technical encryption and user comprehension is emphasized by Miller (2019).

The false impression of complete security on Telegram is caused by the assumption that all conversations are encrypted from beginning to end. The ethical necessity of clear, user-friendly communication is crucial to ensure informed consent.

Ethical Frameworks for Engineers:

Harris, Pritchard, and Rabins (2019) demonstrate that deontological ethics prioritize duties such as honesty, avoiding harm, or autonomy, while utilitarianism emphasizes the importance of consequences and overall welfare.

Telegram's trade-off between privacy, scalability, and usability is assessed by these framework.

Professional Codes

The IEEE and ACM codes say that technology should protect user privacy, be clear and open, and focus on public welfare. In Telegram's case, there is a gap between these ideas and how the app really works. For example, full encryption is not always used, and the rules are not explained clearly to users.

Platform Governance Models

According to Brown and Davis (2021), it has been found that it is not easy at all to find the right balance between free speech and blocking harmful content. Although, Telegram gives people a lot of freedom to talk and share, but it is also true that this freedom makes it easier for misinformation to spread. A possible solution can be to combine human checks with AI tools, which will ensure the platform stays open but also safer.

Comparative Analyses

Chen (2018) and Wong (2020) compared Telegram with different messaging apps. It shows, Telegram is designed to be very fast and handle large groups, whereas, Signal puts more focus on privacy for all chats by default (using end-to-end encryption).

Synthesis

The studies clearly show that ethical design in the apps should properly combine strong security with clear and simple communication, that will allow users know exactly how safe their data and messages are. Professional codes and frameworks guide duty and responsibility, but real-world governance must balance privacy, free expression, and operational constraints. Content moderation remains particularly complex, requiring innovative and accountable approaches. These insights form the foundation for evaluating Telegram's ethical decisions and shaping recommendations.

5. Methodology / Approach

This report employs a qualitative case study methodology to explore the ethical challenges in the platform design and governance of Telegram. The approach incorporates secondary data and ethical theory, combining empirical evidence with theoretical analysis. The methodology has four key components:

5.1 Document Analysis

A thorough examination of publicly accessible documents forms the basis of the study. Technical papers, privacy policies, and terms of service describe data handling, as well as research papers and reports from experts on Telegram's security and governance. The study provides an overview of Telegram's design and operation by examining these sources, as well as ethical considerations related to transparency, user acceptance, and privacy.

5.2 Ethical Framework Application

Telegram's choices are evaluated using standard ethical principles:.

Deontological Ethics investigates the extent to which Telegram fulfills its moral responsibilities, including honesty, privacy, and protection from potential harm.

Utilitarian Ethics evaluates choices based on overall outcomes for stakeholders, considering the benefits of scalable features while considering potential harms from partial encryption and minimal moderation.

Both moral responsibilities and practical considerations are balanced by these complementary lenses.

5.3 Incident Analysis

The 2019–2020 Hong Kong protests, is analyzed, which shows the role of Telegram in real world events:

- How platform features supported secure communication and mobilization.
- How encryption defaults and moderation policies affected user safety, including metadata exposure and arrests.
- Broader ethical implications for platform governance and responsibility.

This case provides empirical grounding for the theoretical evaluation.

5.4 Comparative Contextualization

Comparing the design and governance of Telegram with that of WhatsApp and Signal is done in terms of encryption defaults, privacy, transparency (for example when censoring content), and moderation. Benchmarking is used to identify best practices, trade-offs, and ethical principles while assessing feasibility for large-scale platforms.

Summary:

This is a very holistic approach to ethical evaluation, taking into account document analysis and the application of ethical theories, incident review, and cross-platform benchmarking.

This methodology combines theoretical complexity with practical application, creating a solid foundation for actionable ideas.

6. Case Study / Examples

6.1. Hong Kong Protests (2019–2020)

In the Hong Kong protests, people used Telegram a lot. It became the main app for protesters because it let huge groups talk together in a single platform and was really harder for the government to block. Protesters used it to plan where to meet, share news, and send photos and videos without being watched as easily.

However, Telegram's encryption posed risks. Many users mistook for end-to-end encrypted chats, but only "Secret Chats" offered this protection by being activated individually and handily. In regular cloud and group chats, client-server encryption is used to decrypt messages on Telegram's servers. According to reports, officials took advantage of metadata like timestamps and IP addresses from participants to identify them. This serves as a cautionary tale about how the use of technical design strategies to minimize risks in politically sensitive settings.

Telegram's limited moderation of content added complexity to ethical considerations. The promotion of free speech was accompanied by disinformation, hate speech, and doxxing, which targeted demonstrators. By making design and policy choices that both promote political expression but also expose users to significant vulnerabilities, the Hong Kong case highlights the importance of contextual risk assessment and improved ethical safeguards.

6.2. Platform Transparency and Handling of Data

It is very essential for people to be open, as it helps them feel more confident in the app and understand what's truly happening with their data. Telegram does not share as much, so users cannot clearly see the risks.

Telegram’s privacy policy is also hard to read. It uses technical words that many people do not understand. This means normal users cannot easily tell how their data is collected or stored. This is a bigger problem for people who use Telegram in dangerous situations, because they need clear and simple information to stay safe.

7. Ethical Issues and Analysis

The platform design and governance structure of Telegram present a complex ethical landscape involving competing principles and interests of its stakeholders. This section of the report discusses key issues using deontological and utilitarian perspectives, focusing on moral duties and consequences of technical and policy choices.

7.1 Confusing Privacy Settings

Engineers from a deontological perspective are obligated to provide privacy features that respect user autonomy. The ambiguity created by Telegram's client-server encryption for most chats, and the end-to-end encryption review that can only be manually activated through “Secret Chats", is an example of such an obligation not being fulfilled. The majority of users have a false understanding of stronger protection.

Shifting the responsibility to users goes against the principle of non-maleficence, by risking the exposure of sensitive communications, mostly for activists and groups at risk.

Looking at it through the lens of utilitarianism, the troublesome default settings seem to undermine trust on a societal level. People might stop using Telegram or switch to using riskier alternatives, thus raising the risk on a global scale. The benefits in terms of usability and performance simply do not cover the damages these defaults pose, which calls for an ethical review of the blanket security approach.

7.2 Cost vs. Safety

Universal end-to-end encryption carries substantial technical and financial costs, affecting speed, scalability, and infrastructure.

- Utilitarian view: Security, performance, and resources could be optimised for the greatest benefit with a phased or hybrid rollout.
- Deontological view: Protecting user data should be prioritised above all, as users' privacy and safety are most important.

This tension provokes an even larger question about the responsibility of a company, whether it should prioritise efficiency, features, and safeguarding users. Public safety, risk management, and trust in the future should be considered in the ethical budgeting framework rather than short-term profitability.

7.3 Transparency Gaps

Transparency is a fundamental pillar of ethical governance that allows users to make informed decisions and holds the company accountable. Compared with its competitors, Telegram's disclosures are inadequate, particularly in offering information on:

- Compliance with government data requests
- Policies on content moderation and enforcement of the rules
- Security vulnerabilities and their mitigation

This diminishes trust among users as well as external oversight, which violates the ethical principles of the IEEE and the ACM. Regular transparency disclosures would foster responsibility, empower civil society, and enhance confidence among users.

7.4 Content Control Challenges

Telegram's "light-touch" moderation supports free speech and user autonomy, protecting activists in suppressive regimes. However, lesser moderation also allows room for hate speech, extremist content, and doxxing, risking harassment and social harm.

Ethically balanced approaches to control the content include:

- Minimizing unnecessary censorship along with respecting expression
- Detecting harmful content using privacy-preserving AI and human oversight
- Clear reporting options and transparent appeal functionality
- Policies made considering varying risks for different user groups

This approach aligns with duties to prevent harm and utilitarian goals of maximizing well-being of society.

Summary

The ethical challenges faced by Telegram, including ambiguous privacy settings, the trade-off between cost and security, limited transparency, and dilemmas surrounding content moderation, underscore the conflict between safeguarding user rights and ensuring the scalability of services. To effectively address these challenges, it is essential to develop principled yet practical solutions that are informed by professional standards and the tangible impacts on society, thereby fostering trust, safety, and respect for human dignity.

8. Solutions and Recommendations

Based on the ethical issues presented by Telegram, the report identified meaningful recommendations to implement to increase security, transparency, and responsible governance. The recommendations attempt to balance what can be done with what is ethically appropriate in order to provide a path that will help the engineers, platform operators and even the policymakers navigate ethical dilemmas.

8.1 Default End-to-End Encryption

Telegram needs to implement end-to-end encryption (E2EE) by default for all chats, groups and channels if:

- Provides privacy and protection against unauthorized users accessing data.
- Eliminates the confusion attributed to "Secret Chats" and clarifies expectations of security.
- To put users' safety ahead of users' ability to decipher technical complexity

Role out E2EE at scale will be difficult, however innovations of secure multiparty computation and the need for new decentralized key management systems, can and are being implemented as seen with Signal.

8.2 Enhanced Transparency

Telegram needs to regularly publish transparency reports that cover:

- Data requests by the government and reporting about how many of the requests were met with compliance
- Actions to impose content moderation and the eventual outcomes of appeals
- Performed security audits, number of identified vulnerabilities/mitigations

Ultimately - put ethical principles and codes of conduct into practice, to uphold an informed consent and accountability relationship with its users/communities. This aligns well with the adherence to the IEEE and ACM ethical policies.

8.3 User-Centered Privacy Design

Privacy should be as simple as possible for users:

- Default-safe options for the greatest degree of protection
- Clear definitions and contextual tooltips
- Tutorials and guides for privacy-related management

This approach preserves user choice and allows less tech-savvy users to secure their data.

8.4 Independent Ethics Oversight

Telegram should create an independent ethics board composed of engineers, ethicists, attorneys, and human rights defenders to:

- Critique new features and policies before use
- Publish public reports and ethical critique
- Engage users and regulators in an ongoing conversation

This will reinforce consequences and accountability or be seen as acting in the public's interest.

8.5 Scalable Moderation Tools

To accomplish more balance in ensuring privacy and safety of the users, Telegram should:

- Utilize privacy preserving AI to facilitate the detection of harmful content while also maintaining encryption of data.
- Set up moderator onboarding and training with rigorous guiding principles to mitigate over-censorship of information.
- Restore reporting of issues and appeal functionality to ensure fairness and transparency of the platform.

This upholds the safety of the community without risking security concerns of the users.

8.6 Ongoing Ethical Training

The development of ethics in the engineering culture can be accomplished by:

- Conducting workshops on matters such as cybersecurity, privacy, as well as discussion of the social impact of related issues on a regular basis.
- Reinforcing of the IEEE and ACM principles and code of conduct.
- Evaluating performance and development based on significant ethical principles.

This will ensure that the principles regarding privacy, security or liberties are considered at all stages of design, development, deployment, and innovation.

Summary

These recommendations provide a direction to the future value of Telegram grounded in basic ethical principles. By prioritizing privacy and security by default, transparency, user control, independent accountability, and reasonable moderation, Telegram can build trust and credibility, and perhaps even lead the digital platform industry.

9. Conclusion

The development of Telegram highlights the various intricate ethical challenges faced by a number of communication platforms on a world-wide scale. The balancing of privacy, security, cost effective practices, and scalability includes decisions with implications beyond the technical or business rationale. These are deeply moral choices with significant impacts on the society.

Engineers and those who are operating the platforms carry the responsibility to protect user security and privacy as ethical priorities which are not negotiable. Not being able to uphold these duties risks severe consequences, including:

- Loss of trust of user and platform credibility.
- Compromise of individual safety and data privacy.
- Wider societal harms, including but not limited to misinformation, harassment, and censorship.

On the other hand, commitment to ethical engineering creates trust, innovation, and social legitimacy. Platforms that value transparency and accountability, user empowerment, and responsible governance are uniquely capable of upholding free speech and digital rights in a world that is ever-more surveilled and polarized.

Going forward, Telegram and like-mind platforms must embrace thoughtful reflection and improvement to their ethical stance. As technology and regulations change, engineers and regulators must remain flexible, transparent, and accountable. As platforms implement accountability mechanisms, it is also important to keep ethics in mind at every step of design, development, and governance, it is the responsibility of these platforms to be transparent and dependable to enabling secure, open, and responsible digital communication.

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